

CHEN TANG

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EDUCATION

University of California Berkeley

August 2016 - May 2022

PhD in Mechanical Engineering (Control)

Minors: Machine learning, Optimization

Advisor: Prof. Masayoshi Tomizuka

Hong Kong University of Science and Technology

August 2012 - June 2016

BEng in Mechanical Engineering

Minor: Mathematics

Georgia Institute of Technology

August 2014 - December 2014

Exchange Student in Mechanical Engineering

RESEARCH INTEREST

My research interest lies at the intersection of control, robotics, and learning. I aim to develop embodied AI agents operating in human-centered environments. To tackle this research challenge, I am dedicated to exploring principled approaches to integrate data-driven approaches (e.g., deep learning, generative models, reinforcement learning, and imitation learning) with control theory, explainable AI, and causality. My past and current research focused on applications in autonomous driving and robot navigation.

WORK EXPERIENCE

Department of Computer Science, UT Austin

August 2023 - Present

Postdoctoral Fellow (Supervisor: Prof. Peter Stone)

- Working on autonomous driving and robot navigation at LARG.

Department of Mechanical Engineering, UC Berkeley

July 2022 - July 2023

Postdoctoral Scholar (Supervisor: Prof. Masayoshi Tomizuka)

- Leading behavior-related research activities for autonomous driving (e.g., trajectory prediction, motion planning, and control) at Mechanical Systems Control Lab.

Behavior Team, Waymo

May 2021 - August 2021

Intern, Planner Prediction Router ML & Deep Learning (Host: Qiaoqing Yan, Stephane Ross)

- Build a fast conditional prediction model with deep learning based on Multipath, which achieves the same prediction quality as the full model but significantly reduces inference time.

Honda Research Institute US

June 2019 - Dec 2019

Student Research Intern (Mentor: Sujitha Martin)

- Propose an explainable relational inference framework combining IRL and neural relational inference to incorporate domain knowledge into learning models, which enables semantically meaningful explanations of vehicle interaction.

PROGRAMMING SKILLS

Python, C++, PyTorch, Tensorflow, ROS

TEACHING EXPERIENCE

Department of Mechanical Engineering, UC Berkeley

Jan 2020 - May 2020

Graduate Student Instructor - ME 233 Advanced Control System II

– Rating: 4.50/5.00, Department Average: 4.15/5.00

AWARDS AND SCHOLARSHIPS

RSS Pioneers, 2023

ASME DSCD Rising Stars Award, 2022

Graduate Division Block Grant Summer, 2020

IEEE ITSC Best Student Paper Runner-up, 2018

HKUST Academic Achievement Medal (top 1%), 2016

HKUST President's Cup Silver Award, 2016

The Joseph Lau Luen Hung Charitable Trust Scholarship, 2015

The Chartered Institution of Building Services Engineers Prize, 2014, 2015

The Cheng Foundation Scholarships for Chinese Mainland Undergraduate Students, 2014, 2015

HK SAR Government Scholarship Fund Talent Development Scholarship, 2014

ROBOCON Hong Kong Contest First Runner-up, Best Engineering Award, 2013, 2014

ABU ROBOCON Final Eight, Best Engineering Award, 2013

PUBLICATIONS

(The superscript * indicates equal contribution.)

Journal

- J1 **Chen Tang***, Ben Abbatematteo*, Jiaheng Hu*, Rohan Chandra, Roberto Martín-Martín, and Peter Stone. Deep reinforcement learning for robotics: A survey of real-world successes. *Annual Review of Control Robotics and Autonomous Systems*, 2025
- J2 Tommaso Benciolini, **Chen Tang**, Marion Leibold, Catherine Weaver, Masayoshi Tomizuka, and Wei Zhan. Active exploration in iterative gaussian process regression for uncertainty modeling in autonomous racing. *IEEE Transactions on Control Systems Technology (T-CST)*, 2024
- J3 Zhiyu Huang, **Chen Tang**, Chen Lv, Masayoshi Tomizuka, and Wei Zhan. Learning online belief prediction for efficient pomdp planning in autonomous driving. *IEEE Robotics and Automation Letters (RA-L)*, 2024
- J4 **Chen Tang**, Nishan Srishankar, Sujitha Martin, and Masayoshi Tomizuka. Grounded relational inference: Domain knowledge-driven explainable autonomous driving. *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2024
- J5 Catherine Weaver, **Chen Tang**, Ce Hao, Kenta Kawamoto, Masayoshi Tomizuka, and Wei Zhan. BeTAIL: Behavior transformer adversarial imitation learning from human racing gameplay. *IEEE Robotics and Automation Letters (RA-L)*, 2024
- J6 Ce Hao*, Catherine Weaver*, **Chen Tang**, Kenta Kawamoto, Masayoshi Tomizuka, and Wei Zhan. Skill-Critic: Refining learned skills for hierarchical reinforcement learning. *IEEE Robotics and Automation Letters (RA-L)*, 2024
- J7 Wei-Jer Chang*, **Chen Tang***, Chenran Li, Yeping Hu, Masayoshi Tomizuka, and Wei Zhan. Editing driver character: Socially-controllable behavior generation for interactive traffic simulation. *IEEE Robotics and Automation Letters (RA-L)*, 2023
- J8 Jinning Li, **Chen Tang**, Wei Zhan, and Masayoshi Tomizuka. Hierarchical planning through goal-conditioned offline reinforcement learning. *IEEE Robotics and Automation Letters (RA-L)*, 2022

- J9 **Chen Tang***, Zhuo Xu*, and Masayoshi Tomizuka. Disturbance-observer-based tracking controller for neural network driving policy transfer. *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2019
- J10 Ángel Cuenca, Wei Zhan, Julián Salt, José Alcaina, **Chen Tang**, and Masayoshi Tomizuka. A remote control strategy for an autonomous vehicle with slow sensor using kalman filtering and dual-rate control. *MDPI Sensors*, 2019
- J11 Xu Liu, **Chen Tang**, Xiaohan Du, Shuai Xiong, Siyuan Xi, Yuefeng Liu, Xi Shen, Qingbin Zheng, Zhenyu Wang, Ying Wu, et al. A highly sensitive graphene woven fabric strain sensor for wearable wireless musical instruments. *Materials Horizons*, 2017

Conference Proceedings

- C1 Pengcheng Wang*, Chenran Li*, Catherine Weaver, Kenta Kawamoto, Masayoshi Tomizuka, **Chen Tang**, and Wei Zhan. Residual-MPPI: Online policy customization for continuous control. *International Conference on Learning Representations (ICLR)*, 2025 (acceptance rate: 31.24%)
- C2 Yiheng Li*, Seth Z. Zhao*, Chenfeng Xu, **Chen Tang**, Chenran Li, Mingyu Ding, Masayoshi Tomizuka, and Wei Zhan. Pre-training on synthetic driving data for trajectory prediction. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024 (acceptance rate: 47.5%)
- C3 Yixiao Wang, **Chen Tang**, Lingfeng Sun, Simone Rossi, Yichen Xie, Chensheng Peng, Thomas Hannagan, Stefano Sabatini, Nicola Poerio, Masayoshi Tomizuka, and Wei Zhan. Optimizing diffusion models for joint trajectory prediction and controllable generation. In *European Conference on Computer Vision (ECCV)*, 2024 (acceptance rate: 27.9%)
- C4 Yuxin Chen, **Chen Tang**, Ran Tian, Chenran Li, Jinning Li, Masayoshi Tomizuka, and Wei Zhan. Quantifying interaction level between agents helps cost-efficient generalization in multi-agent reinforcement learning. *Reinforcement Learning Conference (RLC)*, 2024 (acceptance rate: 40%)
- C5 Jinning Li, Xinyi Liu, Banghua Zhu, Jiantao Jiao, Masayoshi Tomizuka, **Chen Tang**, and Wei Zhan. Guided online distillation: Promoting safe reinforcement learning by offline demonstration. *IEEE International Conference on Robotics and Automation (ICRA)*, 2024 (acceptance rate: 44.8%)
- C6 Yuxin Chen, **Chen Tang**, Ran Tian, Chenran Li, Jinning Li, Masayoshi Tomizuka, and Wei Zhan. Quantifying agent interaction in multi-agent reinforcement learning for cost-efficient generalization. *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2024 (Extended Abstract, acceptance rate: 45.1%)
- C7 Chenran Li*, **Chen Tang***, Haruki Nishimura, Jean Mercat, Masayoshi Tomizuka, and Wei Zhan. Residual q-learning: Offline and online policy customization without value. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023 (acceptance rate: 26.1%, featured in **Nikkei Robotics**)
- C8 Shaoshu Su, Ce Hao, Catherine Weaver, **Chen Tang**, Wei Zhan, and Masayoshi Tomizuka. Double-iterative gaussian process regression for modeling error compensation in autonomous racing. *IFAC World Congress (IFAC)*, 2023
- C9 Chenfeng Xu*, Tian Li*, **Chen Tang**, Lingfeng Sun, Kurt Keutzer, Masayoshi Tomizuka, Alireza Fathi, and Wei Zhan. PreTraM: Self-supervised pre-training via connecting trajectory and map. *European Conference on Computer Vision (ECCV)*, 2022 (acceptance rate: 28%)
- C10 **Chen Tang**, Wei Zhan, and Masayoshi Tomizuka. Interventional behavior prediction: Avoiding overly confident anticipation in interactive prediction. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022 (acceptance rate: 48%)

- C11 Lingfeng Sun*, **Chen Tang***, Yaru Niu, Enna Sachdeva, Chiho Choi, Teruhisa Misu, Masayoshi Tomizuka, and Wei Zhan. Domain knowledge driven pseudo labels for interpretable goal-conditioned interactive trajectory prediction. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022 (acceptance rate: 48%)
- C12 **Chen Tang**, Wei Zhan, and Masayoshi Tomizuka. Exploring social posterior collapse in variational autoencoder for interaction modeling. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021 (acceptance rate: 26%)
- C13 Jinning Li, **Chen Tang**, Masayoshi Tomizuka, and Wei Zhan. Dealing with the unknown: Pessimistic offline reinforcement learning. *Conference on Robot Learning (CoRL)*, 2022 (acceptance rate: 38.25%)
- C14 Julián M. Salt Ducaju, **Chen Tang**, and Masayoshi Tomizuka. Application specific system identification for model-based control in self-driving cars. *IEEE Intelligent Vehicles Symposium (IV)*, 2020
- C15 **Chen Tang**, Jianyu Chen, and Masayoshi Tomizuka. Adaptive probabilistic vehicle trajectory prediction through physically feasible bayesian recurrent neural network. *International Conference on Robotics and Automation (ICRA)*, 2019 (acceptance rate: 44%)
- C16 Zhuo Xu, Haonan Chang, **Chen Tang**, Changliu Liu, and Masayoshi Tomizuka. Toward modularization of neural network autonomous driving policy using parallel attribute networks. *2019 IEEE Intelligent Vehicles Symposium (IV)*, 2019
- C17 Zhuo Xu*, **Chen Tang***, and Masayoshi Tomizuka. Zero-shot deep reinforcement learning driving policy transfer for autonomous vehicles based on robust control. *International Conference on Intelligent Transportation Systems (ITSC)*, 2018 (**Best Student Paper Runner-up**)
- C18 Jianyu Chen, **Chen Tang**, Long Xin, Shengbo Eben Li, and Masayoshi Tomizuka. Continuous decision making for on-road autonomous driving under uncertain and interactive environments. *IEEE Intelligent Vehicles Symposium (IV)*, 2018
- C19 Cong Zhao, Rui Xu, Kui Song, Dayu Liu, Shuo Ma, **C. Tang**, Chun Liang, Yitshak Zohar, and Yi-Kuen Lee. The capillary number effect on the capture efficiency of cancer cells on composite microfluidic filtration chips. *IEEE International Conference on Micro Electro Mechanical Systems (MEMS)*, 2015

Workshop

- W1 Michael Munje, **Chen Tang**, Shuijing Liu, Zichao Hu, Yifeng Zhu, Jiaxun Cui, Garrett Warnell, Joydeep Biswas, and Peter Stone. SocialNav-SUB: Benchmarking VLMs for scene understanding in social robot navigation. *ICRA 2025 Workshop on Advances in Social Robot Navigation: Planning, HRI, and Beyond and ICRA 2025 Workshop on Human-Centered Robot Learning in the Era of Big Data and Large Models*, 2025
- W2 Zichao Hu, **Chen Tang**, Michael Munje, Yifeng Zhu, Alex Liu, Shuijing Liu, Garrett Warnell, Peter Stone, and Joydeep Biswas. ComposableNav: Instruction-following navigation in dynamic environments via composable diffusion. *ICRA 2025 Workshop on Human-Centered Robot Learning in the Era of Big Data and Large Models*, 2025
- W3 Jiaxun Cui, **Chen Tang**, Jarrett Holtz, Janice Nguyen, Alessandro G Allievi, Hang Qiu, and Peter Stone. Talking vehicles: Cooperative driving via natural language. *AAAI 2025 Workshop on Advancing LLM-Based Multi-Agent Collaboration*, 2025 (**Oral Presentation**)
- W4 Wei-Jer Chang*, **Chen Tang***, Chenran Li, Yeping Hu, Masayoshi Tomizuka, and Wei Zhan. Editing driver character: Socially-controllable behavior generation for interactive traffic simulation. *CVPR Workshop on Multi-Agent Behavior: Properties, Computation, and Emergence (MABe)*, 2023

W5 **Chen Tang**, Nishan Srishankar, Sujitha Martin, and Masayoshi Tomizuka. Explainable autonomous driving with grounded relational inference. *NeurIPS Workshop on Machine Learning for Autonomous Driving (ML4AD)*, 2020

Preprints and Working Papers

- P1 Michael Munje, **Chen Tang**, Shuijing Liu, Zichao Hu, Yifeng Zhu, Jiaxun Cui, Garrett Warnell, Joydeep Biswas, and Peter Stone. SocialNav-SUB: Benchmarking VLMs for scene understanding in social robot navigation. 2025, under double-blind review
- P5 Zichao Hu, **Chen Tang**, Michael Munje, Yifeng Zhu, Alex Liu, Shuijing Liu, Garrett Warnell, Peter Stone, and Joydeep Biswas. ComposableNav: Instruction-following navigation in dynamic environments via composable diffusion. 2025, under double-blind review
- P2 Yiheng Li, Cunxin Fan, Chongjian Ge, Zhihao Zhao, Chenran Li, Chenfeng Xu, Huaxiu Yao, Masayoshi Tomizuka, Bolei Zhou, **Chen Tang**, Mingyu Ding, and Wei Zhan. WOMD-Reasoning: A large-scale language dataset for interactions and driving intentions reasoning. *arXiv preprint arXiv:2407.04281*, 2025, under double-blind review
- P3 Yuxin Chen*, **Chen Tang***, Chenran Li, Ran Tian, Peter Stone, Masayoshi Tomizuka, and Wei Zhan. MEReQ: Max-ent residual-q inverse rl for sample-efficient alignment from human intervention. *arXiv preprint arXiv:2406.16258*, 2024, under double-blind review
- P4 Ce Hao, **Chen Tang**, Eric Bergkvist, Catherine Weaver, Liting Sun, Wei Zhan, and Masayoshi Tomizuka. Outracing human racers with model-based planning and control for time-trial racing. *arXiv preprint arXiv:2211.09378*, 2022

INVITED TALKS

Learning and Control for Human-Centered Autonomy

- Intelligent Control Lab, Robotics Institute, Carnegie Mellon University, 2025
- Control and Robotics Lab, Department of Mechanical Engineering, Texas A&M University, 2025
- Control Seminar, Department of Mechanical Engineering, University of California Berkeley, 2025
- Department of Mechanical Engineering, The University of Texas at Austin, 2025
- Department of Civil and Environmental Engineering, University of California Los Angeles, 2025
- Robotics Engineering Colloquium Speaking Series, Worcester Polytechnic Institute, 2025
- Department of Mechanical Engineering, Colorado State University, 2025
- Department of Intelligent Systems Engineering, Indiana University Bloomington, 2025

Towards Trustworthy Human-Centered Autonomy

- Safe AI Lab, Department of Mechanical Engineering, Carnegie Mellon University, 2024
- Mitsubishi Electric Research Laboratories, 2024

Diagnosing Social Interaction Modeling for Interactive Autonomous Driving

- Workshop on Social, Interactive and Safe Behaviors for AVs: Benchmarks, Models, and Applications at IEEE Intelligent Vehicles Symposium (IV), 2023

Designing Explainable Autonomous Driving System for Trustworthy Interaction

- ASME DSCD Rising Stars Invited Talks, 2022
- Robotics: Science and Systems Pioneers Workshop, 2023

Exploring Social Posterior Collapse in VAE for Interaction Modeling

- ASL Lab Seminar, Department of Aeronautics and Astronautics, Stanford University, 2021
- Intelligent Control Lab, Robotics Institute, Carnegie Mellon University, 2021

Zero-shot Deep RL Driving Policy Transfer for Autonomous Vehicles based on Robust Control

- Workshop on Reinforcement Learning for Transportation at IEEE Intelligent Transportation Systems Conference (ITSC), 2018

SERVICES AND OUTREACHING

Journal Reviewer

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transaction on Automation Science and Engineering (T-ASE)
- IEEE Transaction on Intelligent Transportation Systems (T-ITS)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)
- Transportation Research: Part C (TR-C)

Conference Reviewer

- AAAI Conference on Artificial Intelligence (AAAI)
- American Control Conference (ACC)
- Conference on Computer Vision and Pattern Recognition (CVPR)
- Conference on Neural Information Processing Systems (NeurIPS)
- Conference on Robot Learning (CoRL)
- IEEE International Conference on Control, Automation, Robotics, and Vision (ICARCV)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE Intelligent Transportation Systems Conference (ITSC)
- IEEE Intelligent Vehicles Symposium (IV)
- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- International Joint Conference on Artificial Intelligence (IJCAI)
- Learning for Dynamics and Control (L4DC)
- Modeling, Estimation, and Control Conference (MECC)
- North American Manufacturing Research Conference (NAMRC)
- Robotics: Science and Systems (RSS)
- World Congress of the International Federation of Automatic Control (IFAC)

Program Committee

- Associate Editor at IEEE Intelligent Transportation Systems Conference (ITSC), 2021-2024
- Lead Organizer of IEEE ICRA Workshop on Human-Centered Robot Learning in the Era of Big Data and Large Models, 2025
- Lead Organizer of NeurIPS Workshop on Progress and Challenges in Trustworthy Embodied AI, 2022
- Co-organizer of IROS Workshop on Multi-Agent Interaction and Relational Reasoning, 2021
- Co-organizer of IEEE IV Workshop on Development of Socially-Compliant Driving Behaviour for Automated Vehicles to Enhance Safety and Efficiency in Mixed Traffic, 2023
- Program Committee of NeurIPS Workshop on ML for Autonomous Driving (ML4AD), 2022
- Program Committee of NeurIPS Workshop on Robot Learning, 2023
- Program Committee of Symposium on ML for Autonomous Driving, 2023
- Program Committee of ICCV Workshop on Visual Perception for Navigation in Human Environments: The JackRabbit Human Motion Forecasting, 2023
- Program Committee of RSS/Pioneers, 2024
- UT Austin Computer Science Ph.D. Admission Committee

Outreaching

- Volunteering mentor in the STEM Muse Mentorship Program

MENTORSHIP

Undergraduate Students

- Lu Wen (now a Ph.D. student at the University of Michigan)
- Tianchen Ji (now a Ph.D. student at UIUC)
- Chen Liu (now a Ph.D. student at the University of Cambridge)
- Yiping Dong (now a software engineer at DeepRoute.ai)
- Chenran Li (now a Ph.D. student at UC Berkeley)
- Vade Shah (now a Ph.D. student at UC Santa Barbara)
- Pengcheng Wang (now a Ph.D. student at UC Berkeley)
- Haotian Lin (now a Master's student at CMU)
- Anthony Wang (undergraduate student at UT Austin)

Master Students

- Julián Salt Ducaju (now a Ph.D. student at Lund University)
- Eric Bergkvist (now an application engineer at Embotech)
- Yaru Niu (now a Ph.D. student at CMU)
- Shaoshu Su (now a Ph.D. student at the University at Buffalo)
- Zhihao Zhao (now a Ph.D. student at UCLA)
- Xinyi Liu (Master's student at the University of Michigan)

PhD Students

- Jinning Li (Ph.D. student at UC Berkeley)
- Ce Hao (Ph.D. student at UC Berkeley)
- Catherine Weaver (Ph.D. student at UC Berkeley)
- Lingfeng Sun (Ph.D. student at UC Berkeley)
- Chenfeng Xu (Ph.D. student at UC Berkeley)
- Wei-Jer Chang (Ph.D. student at UC Berkeley)
- Yiheng Li (Ph.D. student at UC Berkeley)
- Chenran Li (Ph.D. student at UC Berkeley)
- Yuxin Chen (Ph.D. student at UC Berkeley)
- Yixiao Wang (Ph.D. student at UC Berkeley)
- Tommaso Benciolini (Ph.D. student at TUM)
- Michael Munje (Ph.D. student at UT Austin)
- Zichao Hu (Ph.D. student at UT Austin)